



Faculty of Computing

**SECP2523 – DATABASE
SEM 1 - 2025/ 2026**

**INITIAL SYSTEM
PROPOSAL**

Hasta Car Rental System

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To ensure legibility, high-resolution versions of all figures and diagrams included in this report can be accessed via this [Google Drive link](#):  [High-Resolution Report Photos](#)

Table of Content

Glossary.....	3
1.0 Introduction.....	5
2.0 Background Study.....	6
2.1 Existing System Limitations.....	6
2.2 Database Management in Car Rental Systems.....	7
3.0 Problem Statement.....	8
4.0 Proposed Solutions.....	9
4.1 Technical Feasibility.....	9
4.2 Operational Feasibility.....	9
4.3 Financial Feasibility.....	9
4.4 Time Feasibility.....	10
5.0 Objectives.....	11
6.0 Scope.....	12
6.1 User Management Module.....	12
6.2 Vehicle Management Module.....	12
6.3 Booking Module.....	13
6.4 Payment Module.....	14
6.5 Reporting Module.....	14
6.6 Boundaries of Database Application.....	15
6.7 Major User Views.....	15
7.0 Project Planning.....	16
7.1 Human Resources.....	16
7.2 Work Breakdown Structure (WBS).....	17
7.3 Gantt Chart.....	18
8.0 Requirement Analysis (based from AS-IS analysis).....	19
8.1 Current business process (scenarios, workflow).....	19
8.2 Workflow of To-Be System.....	22
9.0 Transaction requirement (data entry, data update/delete, data queries).....	34
9.1 Module 1: User Module.....	34
9.1.1 Module Description.....	34
9.1.2 Transaction Requirement: Data Entry.....	34
9.1.3 Transaction Requirement: Data Update/Deletion.....	34
9.1.4 Transaction Requirement: Data Queries.....	34
9.1.5 Local Conceptual Data Model (ERD).....	35
9.2 Module 2: Vehicle Module.....	36
9.2.1 Module Description.....	36
9.2.2 Transaction Requirement: Data Entry.....	36
9.2.3 Transaction Requirement: Data Update/Deletion.....	36
9.2.4 Transaction Requirement: Data Queries.....	36

9.2.5 Local Conceptual Data Model (ERD).....	38
9.3 Module 3: Booking Module.....	39
9.3.1 Module Description.....	39
9.3.2 Transaction Requirement: Data Entry.....	39
9.3.3 Transaction Requirement: Data Update/Deletion.....	39
9.3.4 Transaction Requirement: Data Queries.....	39
9.3.5 Local Conceptual Data Model (ERD).....	40
9.4 Module 4: Payment Module.....	41
9.4.1 Module Description.....	41
9.4.2 Transaction Requirement: Data Entry.....	41
9.4.3 Transaction Requirement: Data Update/Deletion.....	41
9.4.4 Transaction Requirement: Data Queries.....	41
9.4.5 Local Conceptual Data Model (ERD).....	42
9.5 Module 5: Reporting Module.....	43
9.5.1 Module Description.....	43
9.5.2 Transaction Requirement: Data Entry.....	43
9.5.3 Transaction Requirement: Data Update/Deletion.....	43
9.5.4 Transaction Requirement: Data Queries.....	43
10.0 Benefit and Summary of Proposed System.....	44
11.0 Summary.....	47
Bibliography.....	48

Glossary

Term	Definition
Admin	Refers to General Staff of the car rental company.
AS-IS Analysis	The analysis of the current state of the business process and system (the existing manual and fragmented operations).
Blacklist	A status assigned to a customer who has outstanding issues (e.g., unpaid fines, violations) and is temporarily prevented from making new bookings.
CRUD	An acronym that stands for Create, Read, Update, and Delete. These are the four basic functions necessary for persistence storage in any database-driven application.
Driveo	An external system currently used by staff to record booking history and blacklist records.
DSR	Daily Sales Report. A shared Google Sheet (Excel) that is currently used by staff to record booking and payment details for accounting.
ERD	Entity-Relationship Diagram. A data model illustrating the relationships between entities (tables) in the database.
Feasibility Study	An analysis that examines whether the proposed system is viable in terms of Technical, Operational, Financial, and Time resources.
Gantt Chart	A project management tool used to illustrate the project schedule, showing the start and finish dates of work packages across a timeline.
Hasta Car Rental System	The name of the proposed web-based car rental system to be developed.
Local Conceptual Data Model	The Entity-Relationship Diagram (ERD) specifically designed for a single module within the proposed system.
Loyalty Stamp	A reward unit automatically assigned to customers for completed rentals of a minimum duration (e.g., 3 hours or more), which can accumulate towards a reward or discount.
MySQL	The Relational Database Management System (RDBMS) chosen for the back-end to store and manage the system's data.
RDBMS	Relational Database Management System. A program that allows you to create, update, and administer a relational database.

Term	Definition
Scope	The boundaries and detailed functions included within the design and development of the proposed system.
Stakeholder	An individual or group (in this document referring to En. Alif Firdaus bin Jalaludin) with an interest or concern in the success of the project.
Top Management/ Manager	Refers to Top Management of the car rental company. Individuals that have the same access as Admin but with extra access to reporting and reviewing claims.
Transaction Requirement	Details the necessary data operations (Entry, Update/Deletion, Queries) required for each system module.
UTM	Universiti Teknologi Malaysia. The primary customer base (students and staff) for the car rental service.
WBS	Work Breakdown Structure. A hierarchical decomposition of the total scope of work to be carried out by the project team to accomplish the project objectives.

1.0 Introduction

Hasta Travel & Tours (Sdn. Bhd.) is a car rental company based at Universiti Teknologi Malaysia (UTM), Johor Bahru. It serves mainly UTM students and staff. Currently, the company's daily operations heavily rely on manual record-keeping and communication through WhatsApp. This results in slow booking confirmations, inconsistent vehicle availability updates as well as difficulties in maintaining accurate customer records.

To address these issues, this project aims to develop an integrated web-based car rental system supported by a MySQL database. The database will store and manage operational data such as customer profiles, vehicle inventory and booking details with payment transactions in an organised and structured manner. To ensure data accuracy and integrity, this project uses a relational database management system (RDBMS) to support efficient querying and reporting. This helps prevent double booking caused by availability information not being updated in real time.

The system will be developed using HTML, CSS, PHP and MySQL. With booking, payment, and vehicle return records stored in one centralized database, both staff and customers are able to access accurate and up-to-date information.

2.0 Background Study

2.1 Existing System Limitations

Currently, the booking process is spread across different platforms. Customers contact staff through WhatsApp to ask about vehicle availability and staff then manually key in the booking and driver details into Driveo application. Since most of the work is done manually, the overall process is inefficient and hard to manage:

1. **Lengthy Booking Process:** Customers need to communicate with staff through WhatsApp to check vehicle availability and all bookings are manually confirmed. This causes the booking process to be slow and increases the possibilities of errors when recording booking information.
2. **Redundant Data Entry:** Staff are required to enter the booking details into multiple platforms which include the physical whiteboard, the Wahdah system, and the Excel Daily Sales Report. This repeating process increases staff workload and the risk of transcription errors.
3. **Manual Document Management:** Payment receipts and legal documents are handled manually. Staff need to download, rename and save these files in a specific folder on the computer. This makes document organisation time-consuming and prone to mistakes.
4. **Manual Penalty and Cost Calculations:** Staff must manually calculate and key in financial penalties such as fuel shortages (e.g., RM10 per bar) and late returns into Excel. Since the calculations are not automated, the process depends heavily on staff accuracy and may lead to inconsistent charges.
5. **Separated Operational Tools:** WhatsApp is used for customer communication, Driveo is used to record booking history, and Excel is used for accounting purposes. Staff need to switch between these platforms to complete a single booking, which reduces efficiency.
6. **Disconnected Blacklist Verification:** Although blacklist records exist in the Driveo application, they are not integrated with the WhatsApp booking process. Staff must manually verify each customer's blacklist status during conversations. If this step is missed, bookings may be approved for customers who were previously blacklisted.
7. **Lack of Demographic Insights (Faculty and College):** The system does not categorize bookings based on student demographics, such as Faculty or Residential College. As a result, management is unable to generate summaries to identify which groups of students rent vehicles most frequently.

2.2 Database Management in Car Rental Systems

Database management is significant in car rental systems because it involves a large amount of data to be stored and updated consistently. A Relational Database Management System (RDBMS), such as MySQL is suitable for handling data related to bookings, customers, vehicles, transactions, and blacklisted users in an organized manner. In the proposed system, these data will be stored in separate tables that are linked to each other. This allows information to be connected and retrieved efficiently.

Using MySQL helps reduce the risk of data redundancy and improve data consistency by maintaining relationships between tables. It also supports real time updates of booking records and vehicle availability. This helps prevent issues such as overbooking. Besides, SQL queries can be used to generate financial reports as well as demographic and statistical reports for colleges and faculties.

3.0 Problem Statement

The current car rental booking system at Hasta Travel & Tours Sdn. Bhd. suffers issues that affect both operational effectiveness and customer experience. Currently, the system relies heavily on manual communication. The customers need to communicate with staff through WhatsApp to check vehicle availability and confirm bookings. This repeated manual communication often causes delay in the booking confirmation process.

Additionally, the lack of a proper vehicle booking platform prevents the company from providing customers with essential information such as real-time vehicle availability and detailed vehicle specifications. This makes it difficult for customers to make informed decisions and compare available options. Currently, blacklist records are stored in the Driveo application, but they are not integrated with the WhatsApp booking process. As a result, staff must manually check each customer's blacklist status during booking conversations. If this step is overlooked, bookings may be approved for customers who were previously blacklisted.

Due to these limitations, there is a need for a structured car rental system with a centralized database. A web-based system developed using HTML, CSS, PHP, and MySQL will centralise booking records, customer information and vehicle data into a single platform. This would help reduce manual work, minimize error due to manual entry and improve efficiency for staff while providing a smoother and more reliable booking experience for customers.

4.0 Proposed Solutions

The Hasta Car Rental System is developed to support the daily operations of car rental services for UTM students and staff. The system focuses on integrating key processes into a single platform to improve overall efficiency. By organizing the rental workflow into several core modules, the system is able to support the full rental process, from user registration and booking to payment and record management, in a more structured manner.

4.1 Technical Feasibility

The proposed solution is a web-based car rental management system utilizing HTML, CSS, PHP, and MySQL. These technologies are commonly used in web application development and are suitable for building a database-driven system. The system follows the Model-View-Controller (MVC) architecture pattern to separate user interface, business logic, and data management, making the system easier to maintain and expand.

The user interface is designed using responsive web design techniques to ensure that the system can be accessed through standard web browsers. MySQL is used as the backend relational database to store and manage data such as customer information, vehicle records, bookings, and transactions. Basic security practices such as password hashing and session handling are applied to protect user data and control system access.

4.2 Operational Feasibility

The proposed system enhances operational efficiency by automating currently manual tasks and enforcing consistent rental policies. The system improves the vehicle inspection process by replacing paper forms with a centralized digital interface, which enables digital record keeping of vehicle conditions and damages. Furthermore, the system can identify and track outstanding fines, enabling the admins to verify payment status before approving future transactions and ensuring strict policy adherence. In addition, the user interface is designed with a focus on usability for web environments, which ensures admins can work more efficiently when processing bookings and maintaining records. This allows admins to shift from manual record-keeping to more critical activities like customer service and fleet maintenance.

4.3 Financial Feasibility

The project is made using open-source technologies, which will minimise software licensing costs. The costs will be limited to standard web hosting and server maintenance. This makes the system cost-effective given the system's compatibility with standard web environments. Besides that, the system also provides long-term financial benefit by reducing the expenses associated with printing paper contracts and forms. In addition, the system is able to accurately track vehicle conditions and allow staff to add fines for damages or late returns. This reduces financial issues caused by manual tracking that often occur because of oversight in the manual tracking system.

4.4 Time Feasibility

The development can be done within the given timeframe by using the Agile methodology, which organizes the process into several stages or sprints. This method includes a continuous testing and debugging phase where it can handle several technical issues without delaying the final delivery. This iterative approach often gains regular feedback and review, ensuring the project meets all requirements and stays on track within the deadline.

5.0 Objectives

The primary objective of this project is to develop a web-based car rental system for booking and management with a centralized database for both customers and admin. This system replaces manual processes that enable customers to check vehicle availability and make booking online. It also allows administrators to manage operations through role-based access control. General staff handle daily operations and fleet management while top management can access all staff functions as well as including financial information. In addition, the system aims to improve accuracy of rental cost calculation and provide clear pricing details to customers. This enhances transparency for customers. The project also aims to enhance customer engagement by allowing them to trace their booking history and loyalty rewards. Overall, the proposed system is designed to improve operational efficiency, data accuracy and service quality for the UTM community.

6.0 Scope

In this system, admin include both general staff and top management, with different access levels assigned based on their roles.

6.1 User Management Module

1. **Registration and Login:** During account registration, customers are required to provide verification details such as NRIC or passport, driving license details to ensure a valid account. After registration, customers can register and log in using their UTM student matriculation number or staff ID.
2. **Blacklist Management:** This system allows admin to flag customers with the reasons for the restriction. At this time, blacklist customers are unable to log in to their account and a message is displayed showing the reason and Hasta company contact details. Hence, blacklisted customers may contact the company to resolve the issue. Account access is only restored after the issue has been resolved and the blacklist status is manually removed by an admin.
3. **Loyalty Tracking:** The system tracks customer rental activity and awards loyalty stamps for approved bookings with a minimum 3 hours rental duration.
4. **Role-Based Access Control:** The system differentiates user access levels. **General staff** will have access to daily operational data and fleet management, where more sensitive financial reports and data will be restricted to top management only.
5. **Staff Claims and Reimbursement:** The system provides a claim form to allow staff to claim reimbursement such as fuel costs and car wash along with a receipt. These claim forms are then reviewed and either approved or rejected by top management. For approved claims, top management records the reimbursement transactions in the system. Staff members are able to view the status and claim histories.

6.2 Vehicle Management Module

1. **Car Listings:** The system will display all available vehicles with its details and specifications, including model, passenger capacity and rental rates. This ensures customers select the vehicle that best fits their needs.
2. **Real-Time Availability:** The system will automatically update the availability status of each vehicle once a booking is confirmed or rejected. This ensures real-time inventory accuracy and prevents double bookings.
3. **Maintenance and Document Tracking:** Staff can manage, track, and update maintenance records as well as road tax and insurance details. The system includes an automated alert mechanism that flags vehicles on the dashboard

when their insurance or road tax is set to expire within 30 days, ensuring that the fleet remains road-legal and safe. Upon expiration, the system restricts the vehicle's availability until the admin updates the new validity period.

4. **Check Availability:** The system will show vehicle availability for the current week, including vehicles that are available or booked. Admin can also filter the availability by date and view the corresponding booking details.

6.3 Booking Module

1. **Search and Real Time Availability:** Customers browse and search available cars based on their selected pickup and drop-off dates and times. The system enforces a minimum rental duration of 1 hour. The system performs a real-time conflict check, ensuring that available vehicles remain active for booking, while booked or maintenance vehicles are visually disabled to prevent selection to maintain full fleet visibility without scheduling conflicts.
2. **Booking Confirmation:** When a booking is made, the booking status is initially marked as pending state awaiting administrative verification. Once approved by staff, the booking is confirmed, availability of the vehicle is automatically updated in real time. Additionally, the system automatically calculates and applies extra delivery charges for non-standard drop-off locations and finally provides customers with a clear and listed cost breakdown before confirmation.
3. **Pickup Form:** Before driving vehicles, customers are required to complete a pickup inspection form for recording the initial state of the vehicle such as fuel level and existing scratches. This inspection form serves as important proof for verifying new damage upon return.
4. **Return Form:** When returning a car, customers are required to complete a digital return form within the system. This involves updating the latest fuel level and uploading real time photos of the vehicle's condition. Feedback is also included in the return form and the critical issues are listed. If any critical issue is selected then the car will be automatically set as "Unavailable" until staff resolve that problem and manually reset it as "Available". This can prevent unsafe vehicles from being rented out as well as ensuring customers safety and maintaining fleet quality at the same time.
5. **Automated Notifications:** Customers receive automated system notification and email updates for all their booking status lifecycle changes such as confirmed, modified and rejected. This ensures customers receive instant and accurate status updates.
6. **Modify Booking:** The system will allow admin to modify confirmed bookings when necessary, such as changes due to vehicle issues. Customers will then be

notified if there are any booking changes through system notifications and email updates.

7. **Reschedule and Cancel Booking:** The system will allow customers to reschedule or cancel their booking at least 24 hours before the scheduled rental time.

6.4 Payment Module

1. **Direct payment integration:** Customers can make payments using a DuitNow QR code or direct bank transfer using the provided account details. After the payment, customers will need to upload their payment receipt, which will be manually verified by staff to confirm the booking.
2. **Voucher Redemption:** Customers are able to redeem vouchers by applying a voucher code during the payment process or through the rewards page. Admin can create and manage vouchers and view voucher redemption records.
3. **Financial transparency:** The system will maintain a detailed transaction history, including rentals, security deposits, and penalty fees to ensure fully visible and clear financial records for both admin and customers.

6.5 Reporting Module

1. **Financial and Staff Payroll Reports:** The system will generate financial reporting by calculating total revenue and the net profit after deducting operational costs such as staff salaries and claimed expenses. This allows top management to review financial performance and better understand the overall financial status of the company.
2. **Booking Trends and Demographics:** The dashboard displays total booking data based on faculty and college, allowing management to observe and identify which customer groups rent frequently.

6.6 Boundaries of Database Application

The boundary of the database application covers all data storage and management required by the User, Vehicle, Booking, Payment, and Reporting modules as illustrated in the use case diagram. The database supports operations such as user registration and authentication, vehicle information management, booking creation and updates, payment and deposit tracking, loyalty reward processing, and report generation.

External services such as online payment gateways, notification delivery mechanisms, and user interface rendering are outside the database boundary. These components interact with the database only through stored transaction records, references, and status updates.

6.7 Major User Views

The database supports three major user views as defined in the use case diagram. Customers interact with the database to manage their bookings, payments, and personal information. Administrative staff access operational data related to vehicles, bookings, payments, and blacklist management. Top management users access summarized and aggregated data through dashboards and financial reports for monitoring and decision-making purposes.

Each user view is implemented through role-based access control to ensure data confidentiality and operational security.

7.0 Project Planning

7.1 Human Resources

This section outlines the project team responsible for the development of the Hasta Car Rental System. The project follows a structured development lifecycle, requiring specific roles to ensure successful delivery within the timeline.

Team Member	Role	Responsibilities
Gwee Zi Ni	Project Manager	Monitor overall project scheduling, creating Gantt Chart, ensuring deadlines are met, and managing scope to avoid delays.
Lee Jia Yee	System Analyst	Conducts the feasibility study, gathers requirements and designs the system architecture.
Michelle Ho Chia Xin	Lead Programmer	Develops the core modules, manages the Database and handles deployment.
Tay Xin Ying	UI/UX Designer	Designs the web interface, ensures responsiveness of the website, and creates user-friendly flows.
Kalaitharan A/L Palanyvelu	Quality Assurance (QA)	Performs black-box testing, identifies bugs and ensures the system meets objectives.

7.2 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) breaks down the Hasta Car Rental System project into smaller and more manageable components. It acts as a foundation for defining the overall project scope. The WBS starts hierarchically with the project phases which are initiation, design, development, testing and deployment. Then, all phases are decomposed into specific deliverables and work packages. This structure helps ensure that all system functions are properly planned and developed. The following figure shows the detailed WBS of the project system.

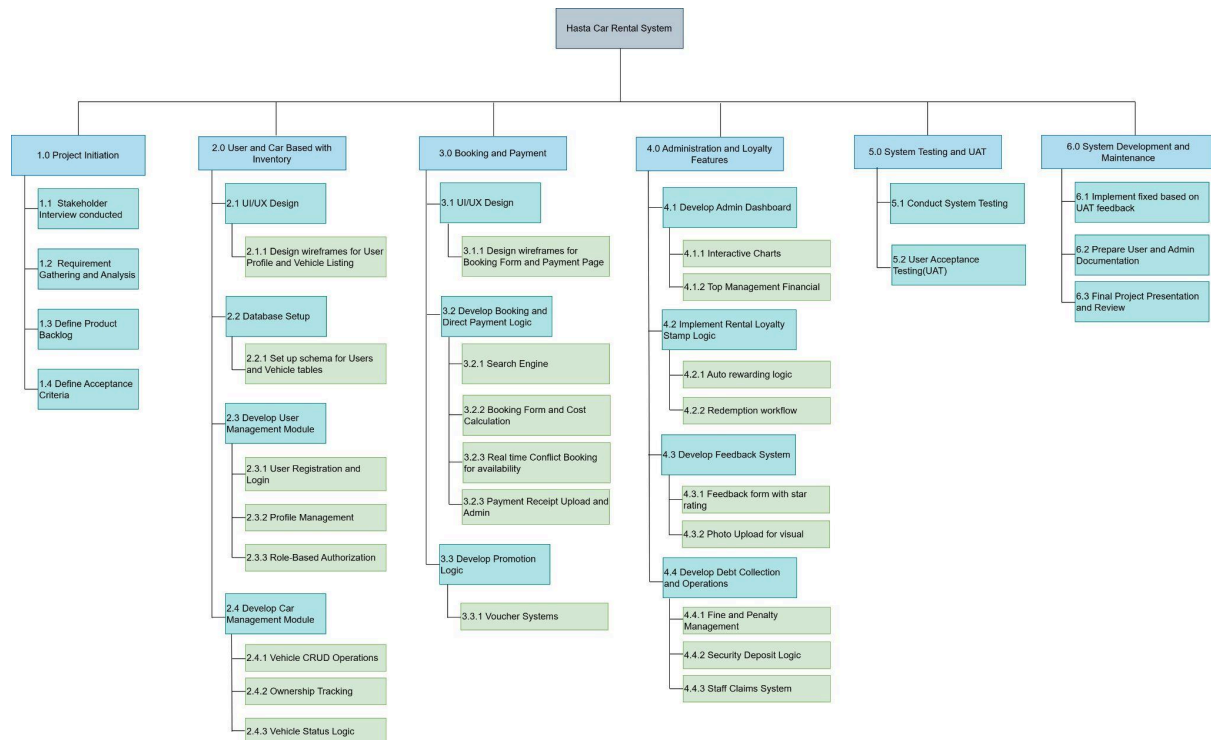


Figure 7.2.1: Work Breakdown Structure(WBS)

7.3 Gantt Chart

Based on the structure defined in the WBS, the project schedule is presented using a Gantt Chart. The chart shows the planned work packages across the 14 weeks project timeline. The development phase follows an Agile approach. There are 3 sprints. Sprint 1 focuses on user management and fleet related features, while Sprint 2 focuses on the main booking and payment functions and Sprint 3 implements advanced administration features such as the loyalty feature and financial reporting. The following figure shows the Gantt Chart with highlighting task durations and the overlap between development activities.



Figure 7.3.1: Gantt Chart for Project Schedule

8.0 Requirement Analysis (based from AS-IS analysis)

8.1 Current business process (scenarios, workflow)

The current business operations for Hasta Travel & Tours depends intensively on manual data entry across multiple platforms: WhatsApp, physical whiteboard, excel and the Wahdah system.

The core business processes are divided into four key scenarios:

Scenario 1: Booking & Reservation Process

This process begins when a customer contacts the company via WhatsApp to rent a vehicle.

1. Inquiry & Verification:

- The customer contacts the staff via WhatsApp.
- The staff sends the Fast Reply Booking Form to let the customer fill in pickup and return details.
- After received reply from the customer, the staff needs to verify three things:
 - Ensure that the customer is UTM student or staff
 - Ensure that pickup and return location is in UTM Student Mall
 - Ensure that the pickup and return time is within office hour (9am - 5pm)

2. Booking Confirmation:

- If the customer chooses a pickup and return location other than UTM Student Mall or chooses pickup and return time outside of office hour: booking can not proceed unless there is a staff that is available to handle this booking. The staff needs to confirm with the staff in charge regarding the additional charge and inform the customer regarding this charge.
- The staff checks the car availability on the physical whiteboard and informs the customer about the available vehicle choices and differences between their prices.
- The staff confirms the vehicle model and the charges with the customer.
- The staff send the Booking Form to let the customer fill in personal details and banking details (for refund), and request the customer to make payment of minimum RM50 as deposit

- After received receipt for deposit payment, the staff needs to request front and back pictures of the customer's documents (IC/Passport, Driving license and matric card)

3. Data Recording:

- **Whiteboard:** Staff writes the booking on the physical whiteboard with different indicators (Black ink for UTM students/staff, Blue for outsiders, Green for HASTA staff, Red means the vehicle unavailable for rental).
- Staff downloads the receipt and saves it in the computer folder.
- **Wahdah System:** Staff manually keys the customer and rental details into the "Wahdah" web system to generate the rental entry.
- **Daily Sales Report (DSR):** Staff enters the booking details into a shared Google Sheet (Excel) named "Daily Sales Report," recording payment details (Cash/Transfer).

Scenario 2: Car Pickup (Handover)

The pickup process has two variations: **Online** and **Physical**.

● A. Online:

1. **System Update:** Staff updates the Wahdah system status to "Paid" if full payment is received.
2. **Document Handling:** Staff downloads the *Agreement Form* and *Inspection Form* as PDFs and sends them to the customer via WhatsApp 30 minutes before pickup.
3. **Digital Verification:** The customer digitally signs the forms and sends them back along with photos of the car (4 sides, mileage, and fuel bar).
4. **Key Handover:** Once verified, the staff instructs the customer to retrieve the key (usually placed under the driver's seat).

● B. Physical:

1. **Printing:** Staff prints the *Inspection Form* and *Agreement Form* in black and white.
2. **Inspection:** Staff and customer physically inspect the car together. Any pre-existing damage is marked on the printed Inspection Form.
3. **Sign-Off:** The customer signs the physical forms. Staff takes photos of the car condition and key handover.

Scenario 3: Car Return & Inspection

This process ensures the asset is returned safely and determines if the security deposit can be refunded.

1. **Tracking:** Staff uses the GPS Tracker (Tracksolid Pro) to verify the return time and location (must be at Student Mall).
2. **Damage & Fuel Check:**
 - Staff compares the car's current condition against the pre-rental photos.
 - Penalties are calculated for fuel shortages (RM10/bar), late returns (charged per hour), or new damage.
3. **System Update:**
 - Staff updates the "Deposit Details" section in the Daily Sales Report (Excel), recording any deductions for fuel or damage.
4. **Completion:** Staff sends a WhatsApp confirmation to the customer stating the deposit will be refunded within 10 working days.

Scenario 4: Staff Administration (Claims & Loyalty)

1. **Loyalty Program:** Staff updates the physical "Loyalty Card" for eligible customers. If a customer rents for 12 hours (1st stamp) or 9 hours (subsequent stamps), they earn progress toward a discount. This is manually tracked in an Excel sheet.
2. **Claims & Commissions:**
 - Staff records "Extra Jobs" (after-hours work) in an Excel file for commission calculation.
 - Operational expenses (petrol, car wash) are recorded in a Claim Excel sheet with attached receipts.

8.2 Workflow of To-Be System

Workflow 1 - Booking Process

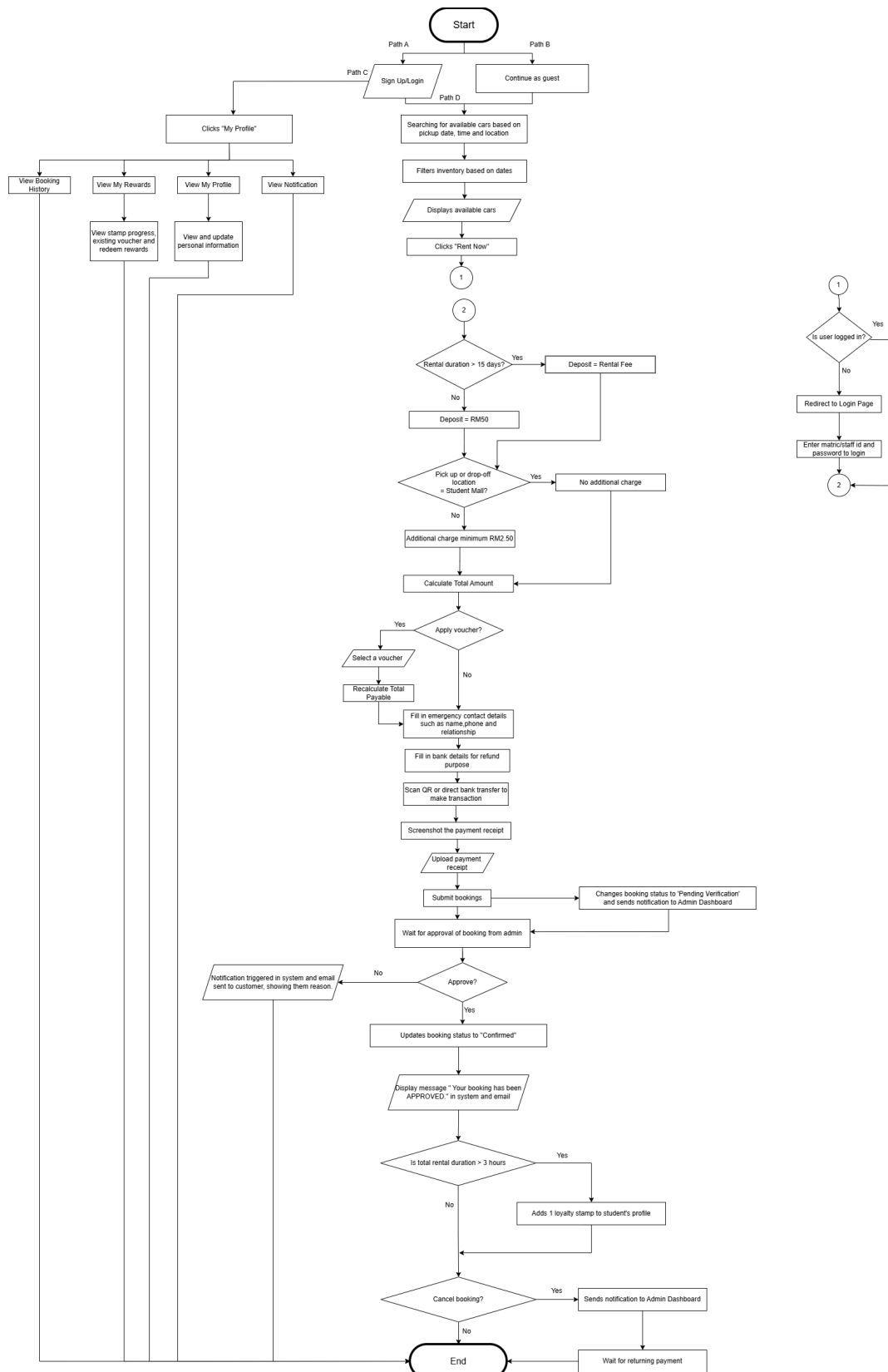


Figure 8.1.1: Workflow 1 for Booking Process

Workflow 2 - Manage Booking

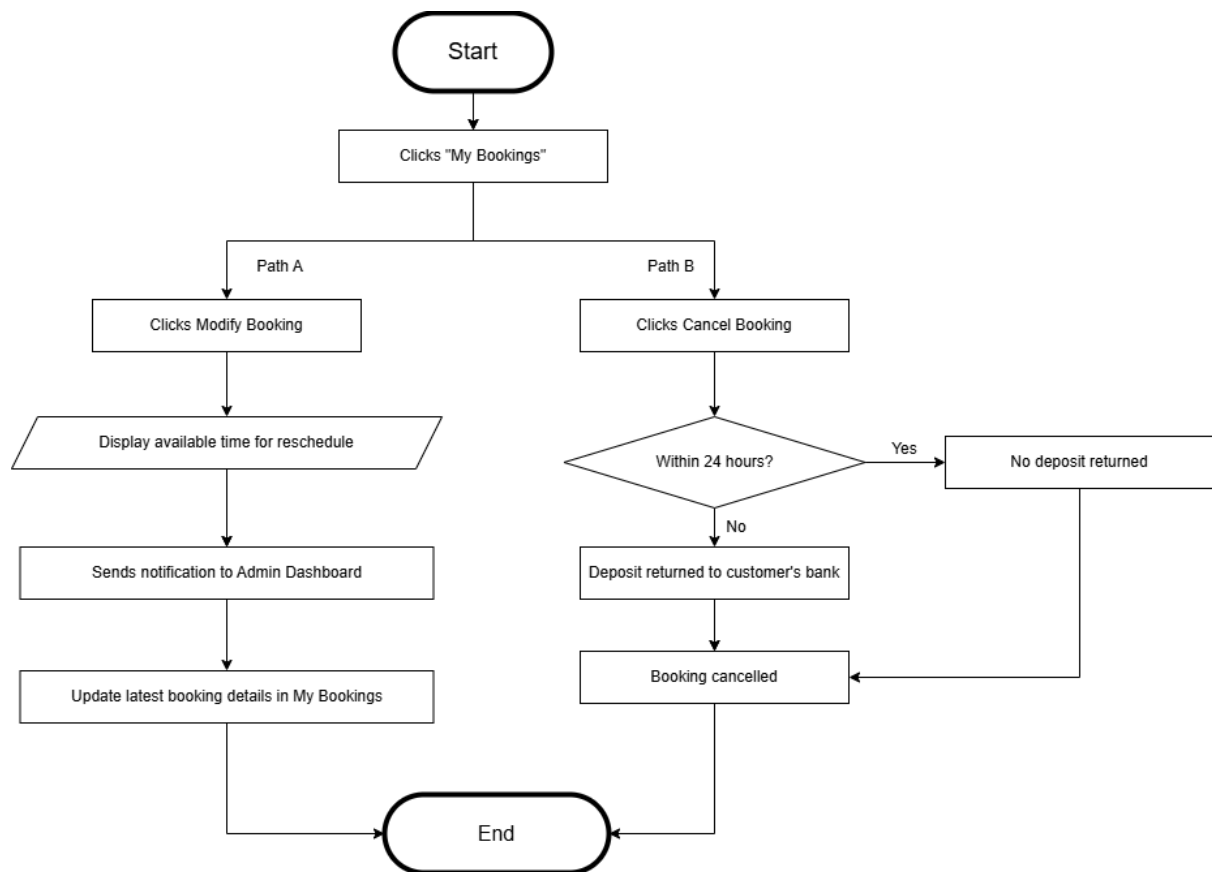


Figure 8.1.2: Workflow 2 for Manage Booking

Workflow 3 - Pick Up Inspection

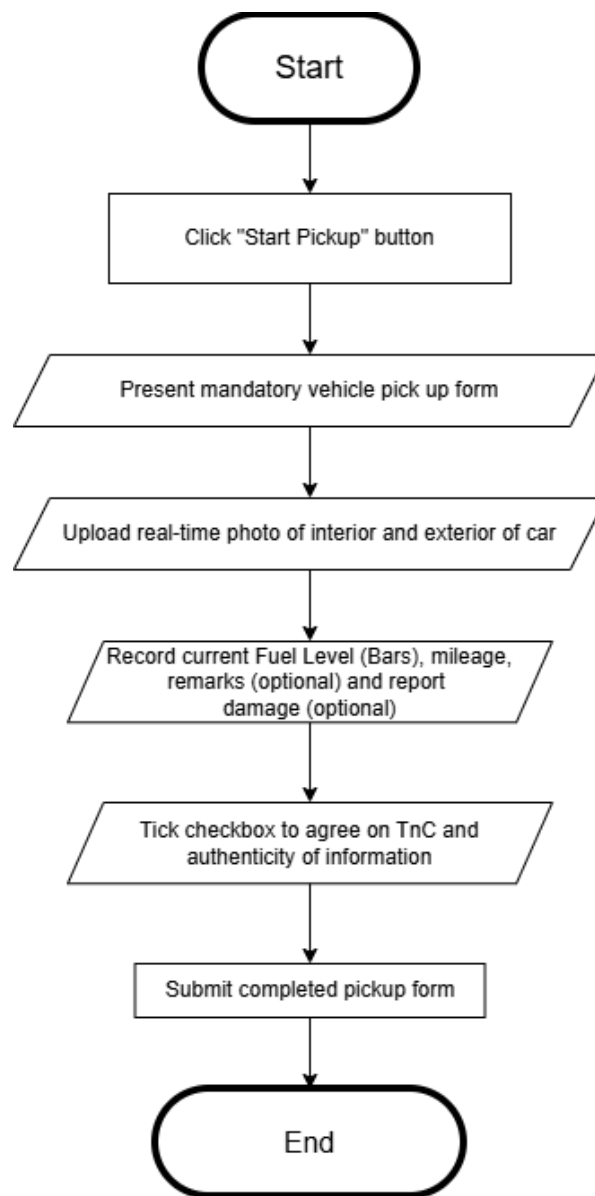


Figure 8.1.3: Workflow 3 for Pick Up Inspection

Workflow 4 - Return Inspection

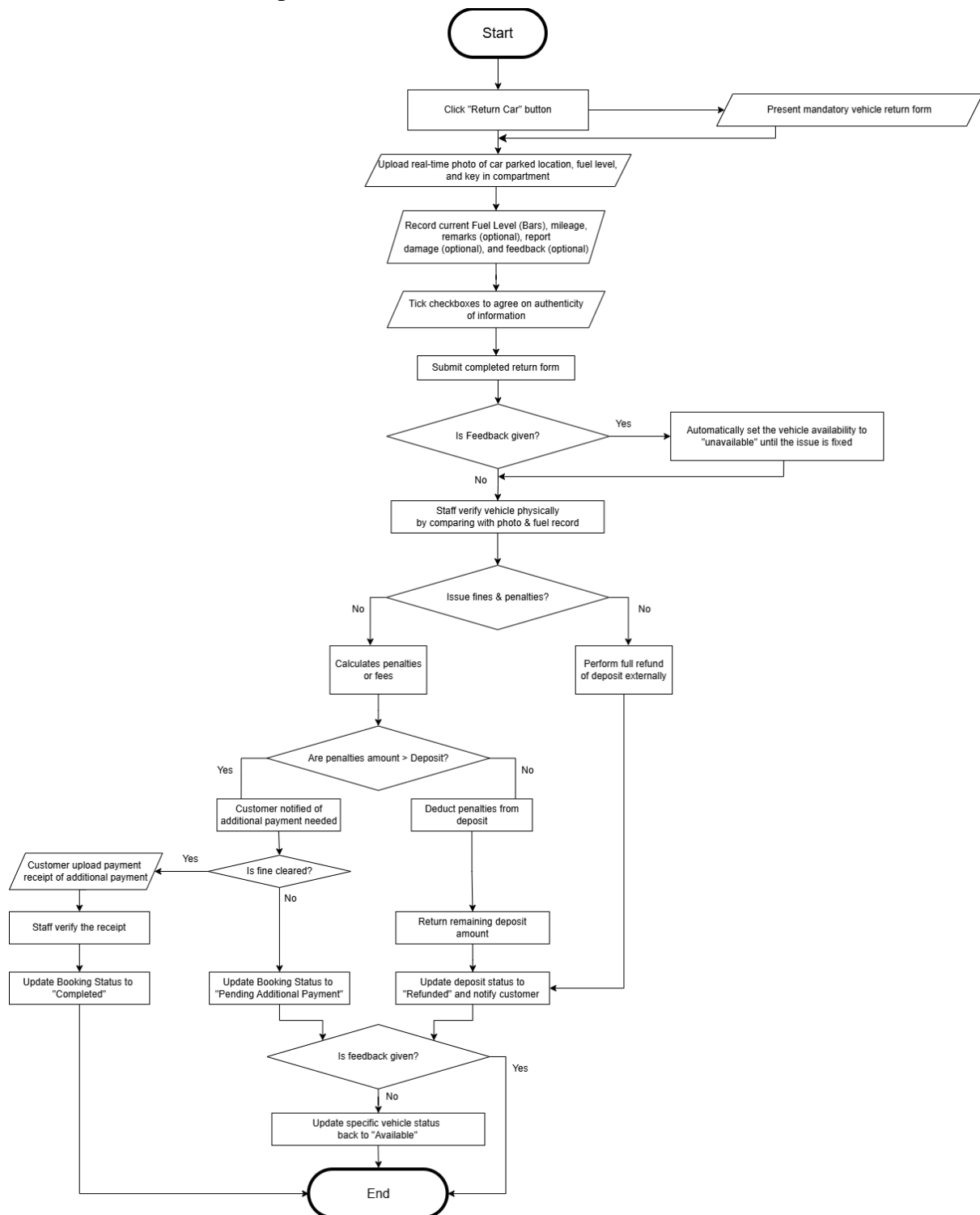


Figure 8.1.4: Workflow 4 for Return Inspection

Workflow 5 - Admin Portal

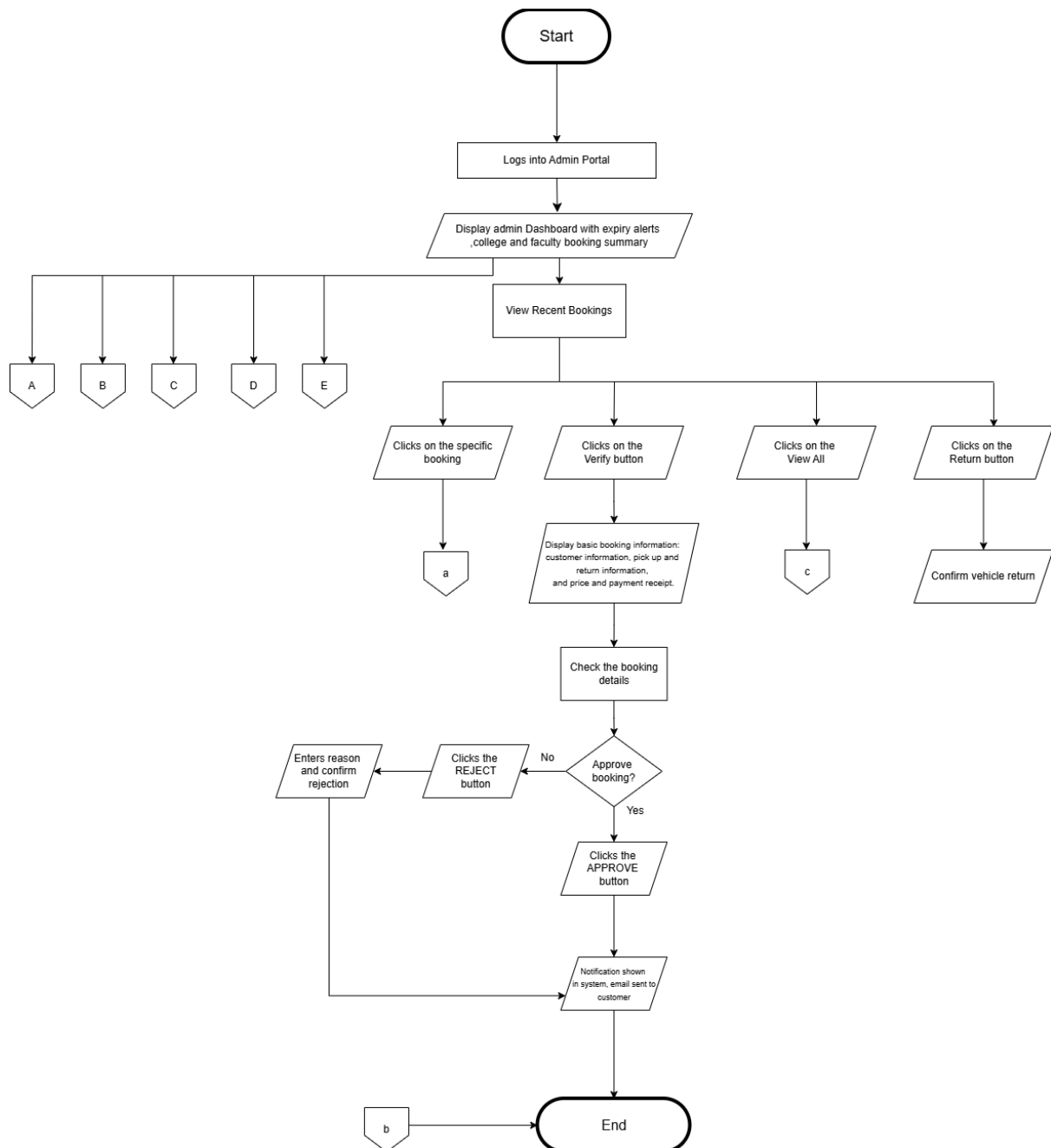


Figure 8.1.5: Workflow 5 for Admin Portal

Workflow 6 - Modify Booking

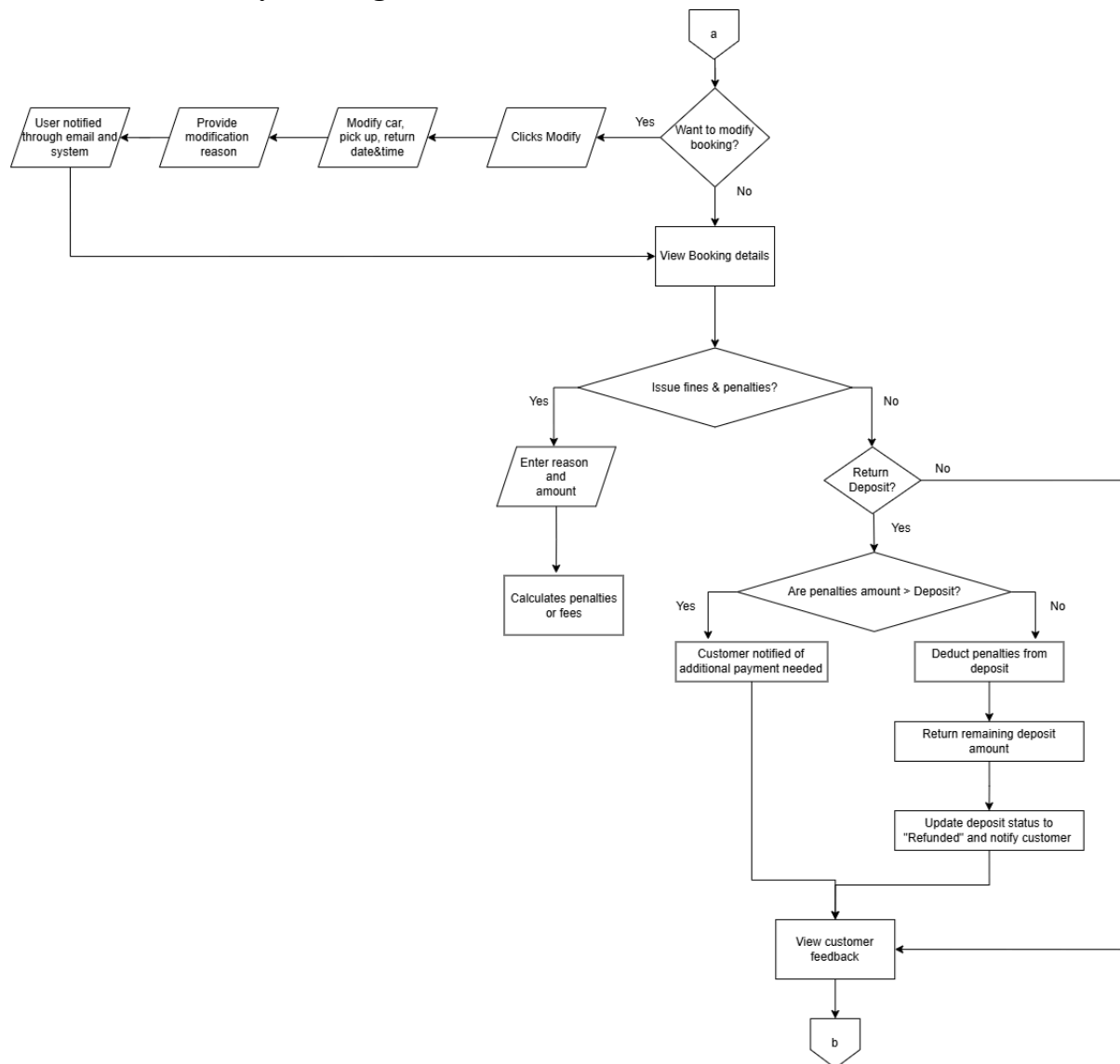


Figure 8.1.6: Workflow 6 for Modify Booking

Workflow 7 - Vehicle Availability

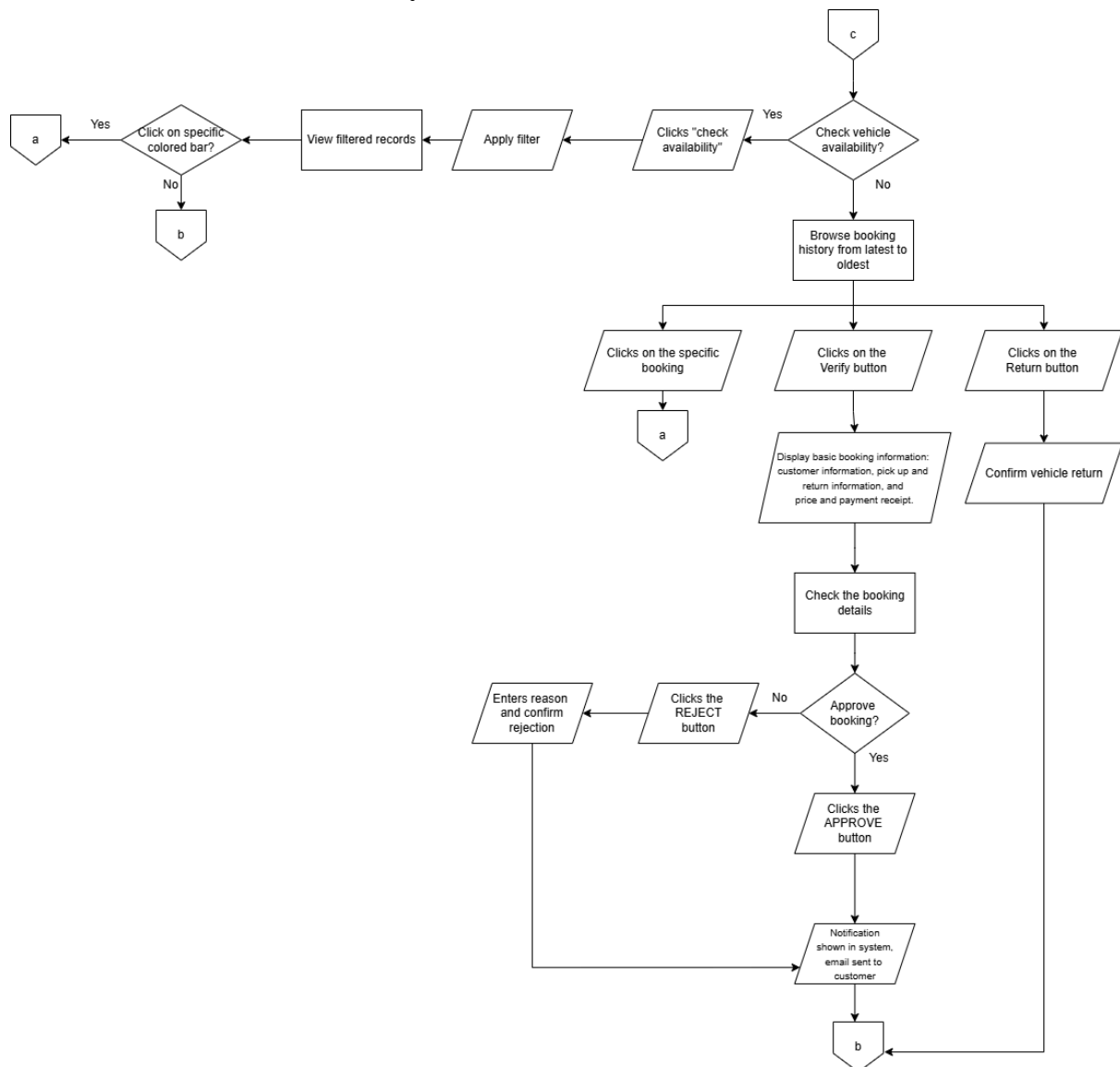


Figure 8.1.7: Workflow 7 for Vehicle Availability

Workflow 8 - Fleet Management

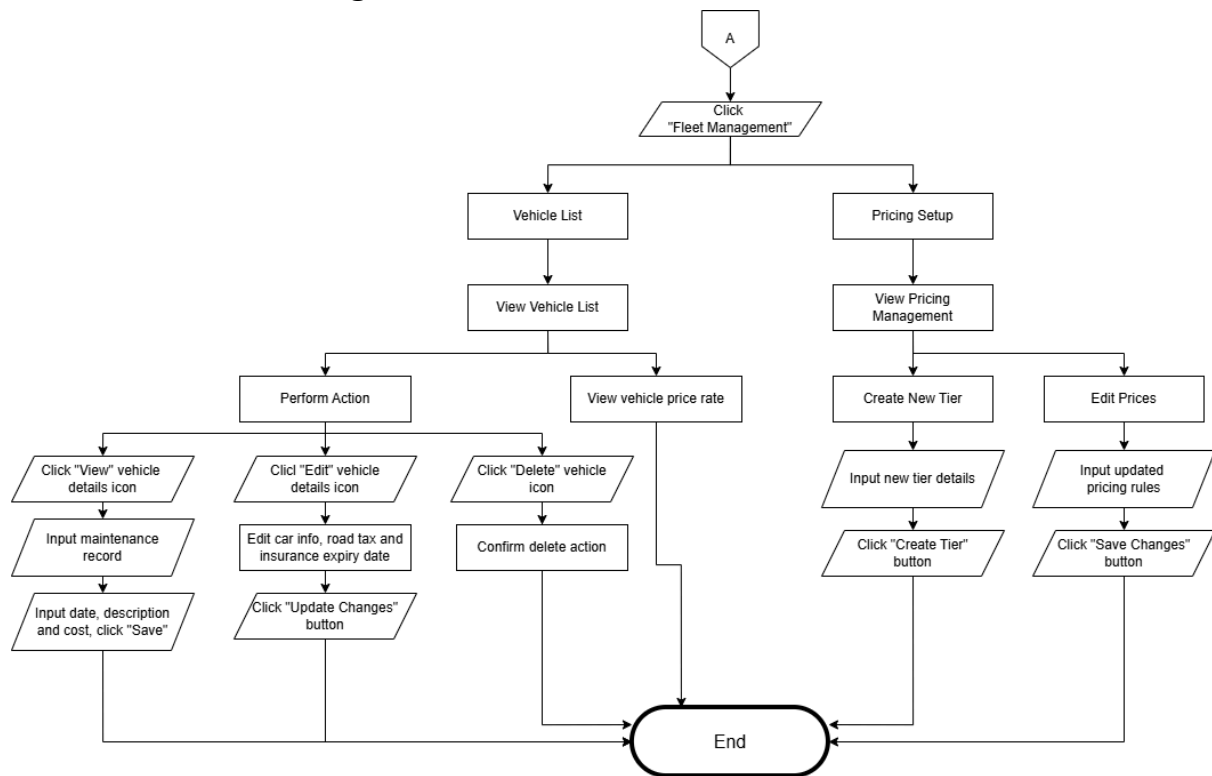


Figure 8.1.8: Workflow 8 for Fleet Management

Workflow 9 - Manage Customer

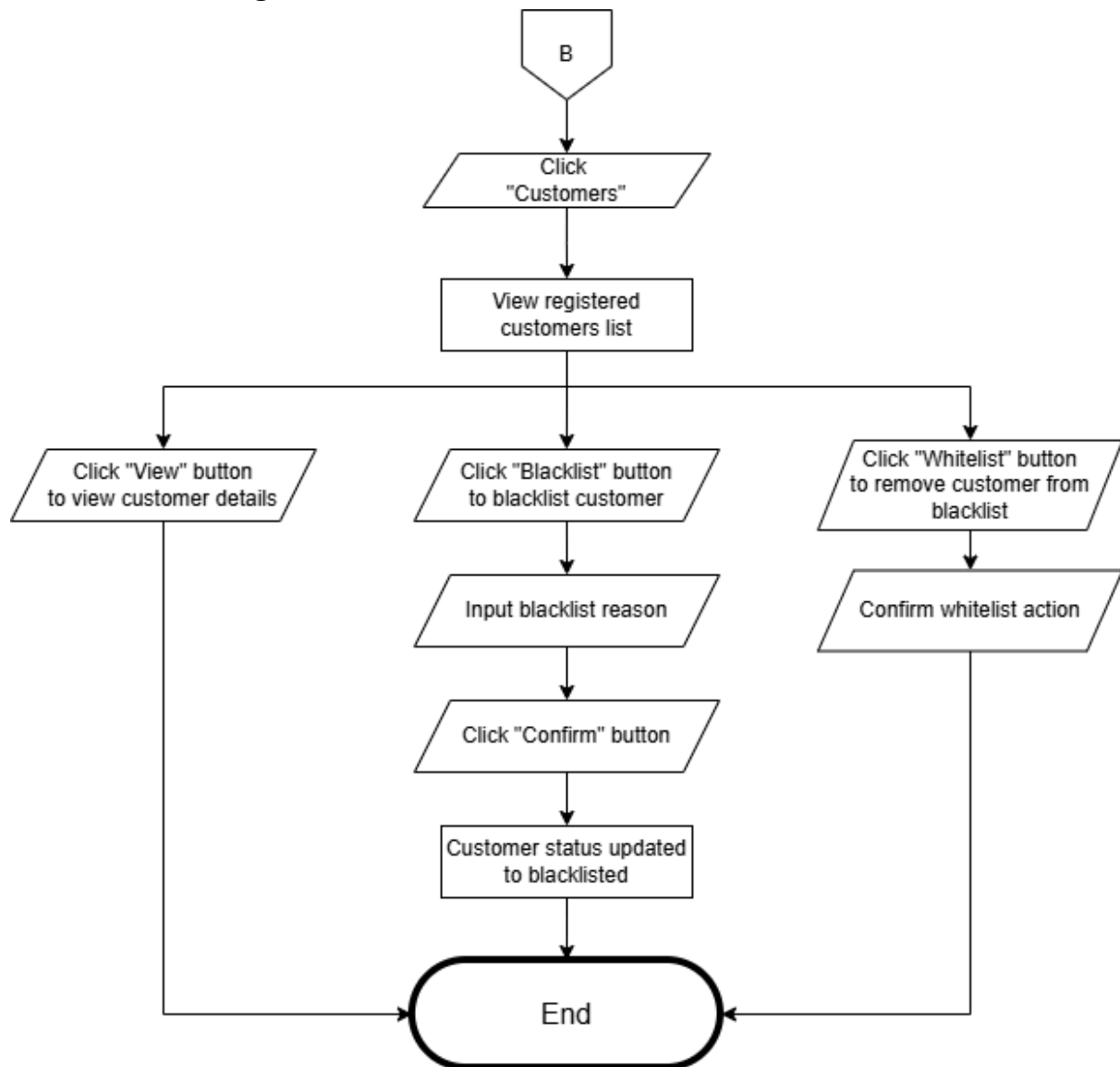


Figure 8.1.9: Workflow 9 for Manage Customer

Workflow 10 - Manage Voucher

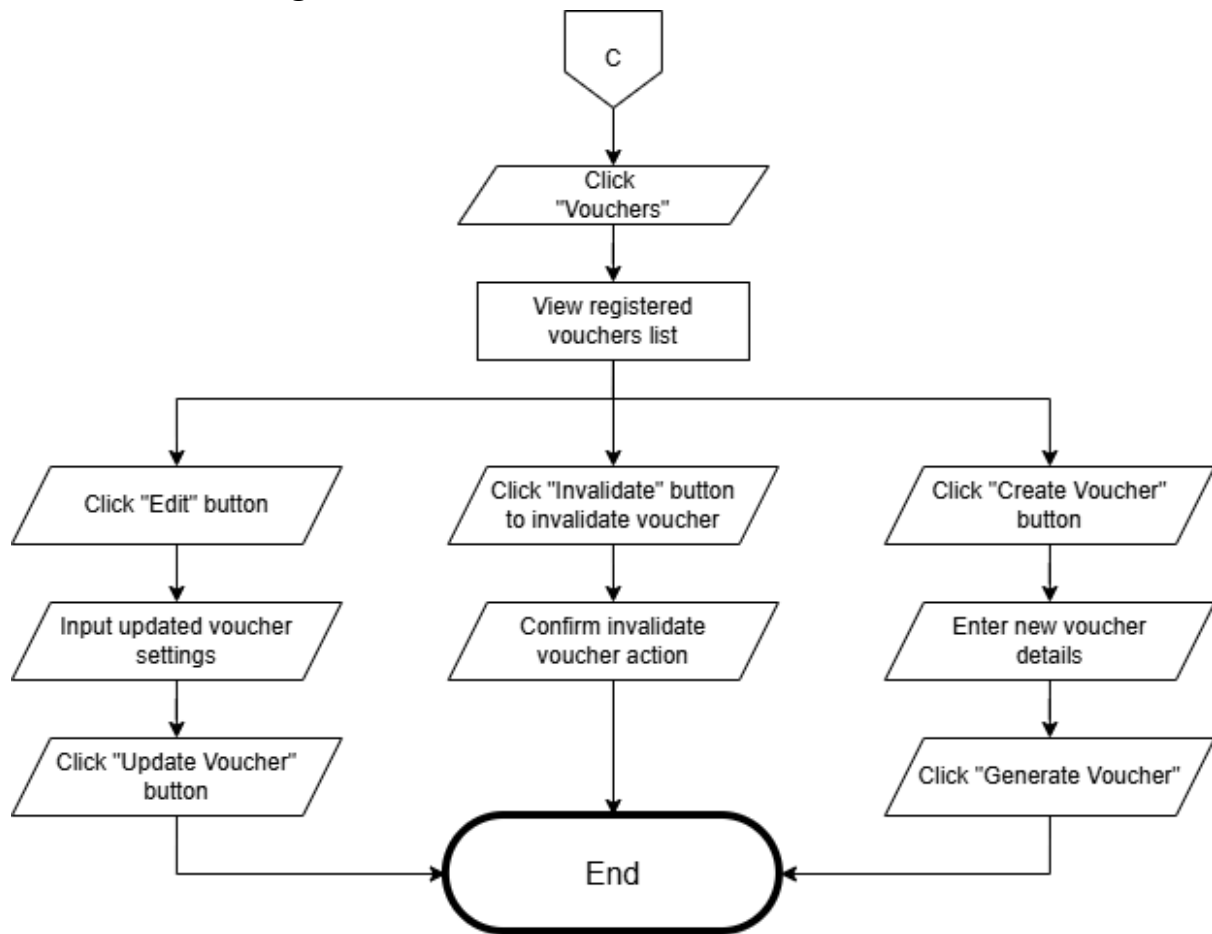


Figure 8.1.10: Workflow 10 for Manage Voucher

Workflow 11 - Manage Feedback

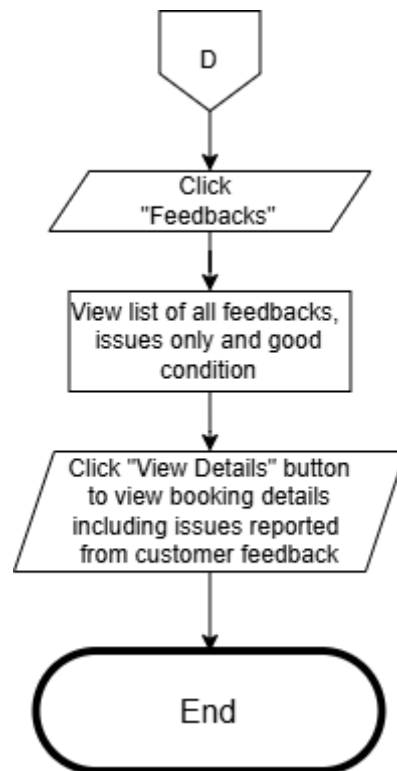


Figure 8.1.11: Workflow 11 for Manage Feedback

Workflow 12 - Claim Form

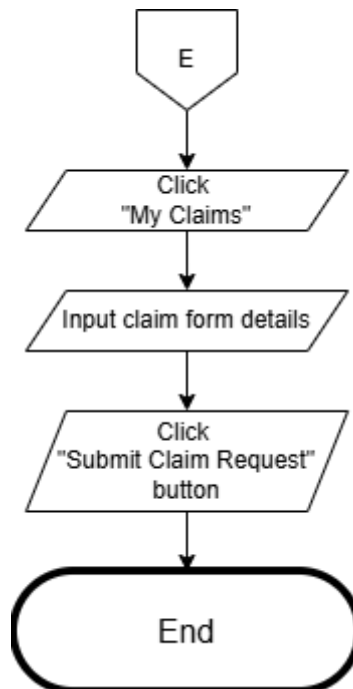


Figure 8.1.12: Workflow 12 for Claim Form

Workflow 13 - Management Portal

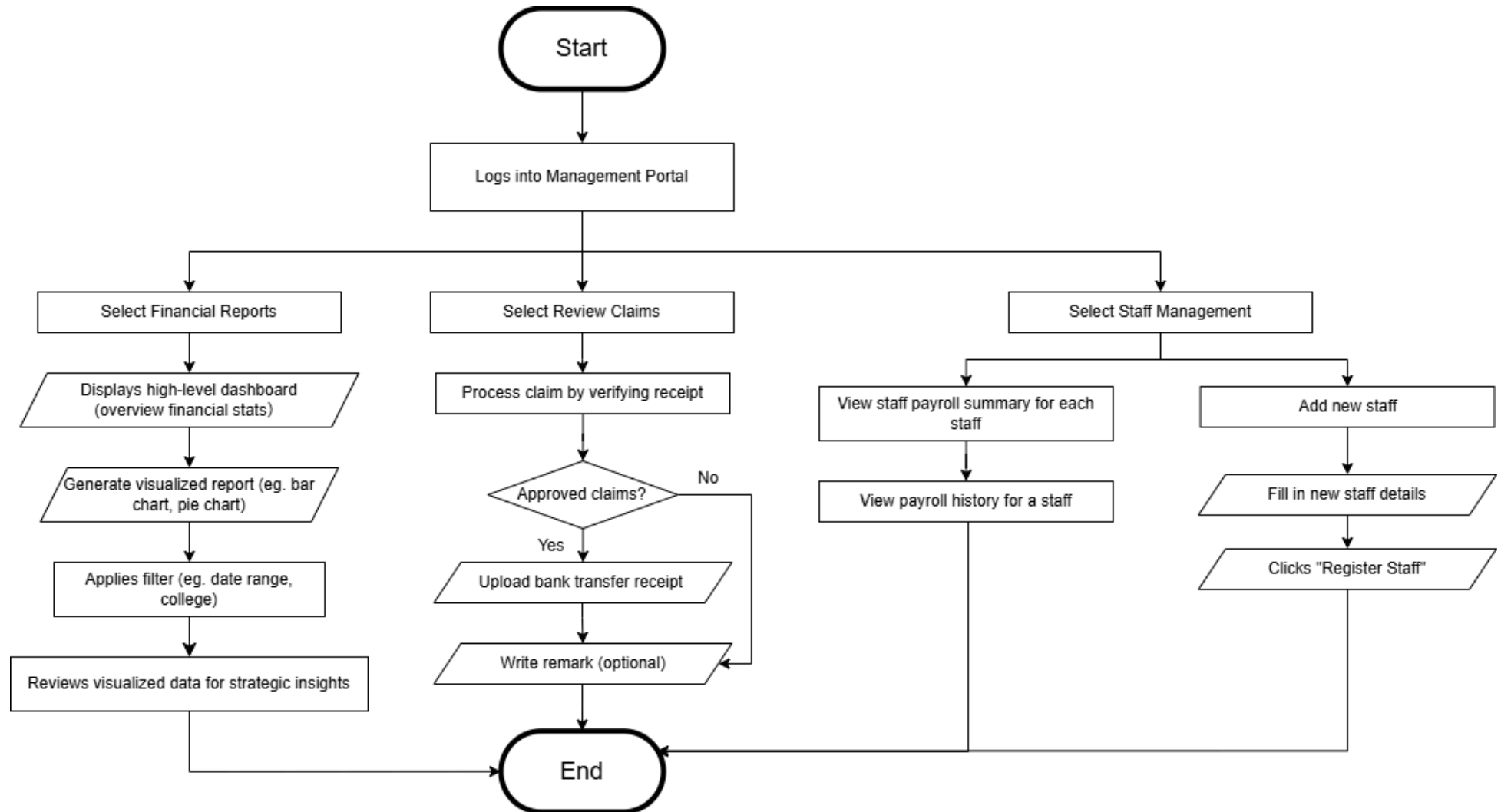


Figure 8.1.13: Workflow 13 for Management Portal

9.0 Transaction requirement (data entry, data update/delete, data queries)

9.1 Module 1: User Module

9.1.1 Module Description

The User Module is a central feature of the Hasta Car Rental System. It allows users (UTM students and staff) to securely register and log in using their student matric number or staff ID. Upon their first login, users will be required to verify their personal details, such as NRIC/Passport and driving license, ensuring the authenticity of the user.

The module also integrates a blacklist system to manage users who have outstanding debts or violations. Blacklisted users are blocked from accessing the system until all problems are resolved. Additionally, the loyalty rewards system tracks users' rental history, allowing customers to view and redeem loyalty stamps after meeting certain conditions, such as accumulated rental hours.

9.1.2 Transaction Requirement: Data Entry

1. Account Registration: Users will enter their personal details, including name, email, matriculation number or staff ID, NRIC/Passport, driving license expiry date, college name and other essential details during registration
2. Login: Customers will log in with their UTM Matric ID/Staff ID, while Hasta's own staff will login via Staff ID, and a password for authentication.

9.1.3 Transaction Requirement: Data Update/Deletion

1. Update Profile: Users can modify their personal information such as email, address, and phone number.
2. Delete Account: Users can request account deletion, which will remove all associated personal data from the system, ensuring complete privacy.

9.1.4 Transaction Requirement: Data Queries

1. List of Users: Query to display all registered users with their username/email, registration date, account status.
2. View Profile: The system queries the database to retrieve the currently logged-in user's personal information, including their NRIC/Passport, contact details, faculty, and college associations, to display on the profile management page.
3. Booking History: The system retrieves a chronological list of all past and active bookings associated with the user's account. This query includes details

of the rented vehicles, rental duration, and payment status to allow users to track their usage history.

4. **Loyalty & Vouchers Status:** A query is executed to fetch the user's current loyalty card status, specifically the number of accumulated stamps. Similarly, the system queries for any active, unused vouchers linked to the user's account for display in the rewards section.
5. **Blacklist Status Check:** During critical actions such as logging in or attempting to make a new booking, the system queries the user's account status to verify if they are currently blacklisted due to outstanding fines or violations, preventing access if necessary.
6. **Customer Search (Admin):** For administrative purposes, the system supports querying the user database to view customer lists and details to facilitate user management.

9.1.5 Local Conceptual Data Model (ERD)

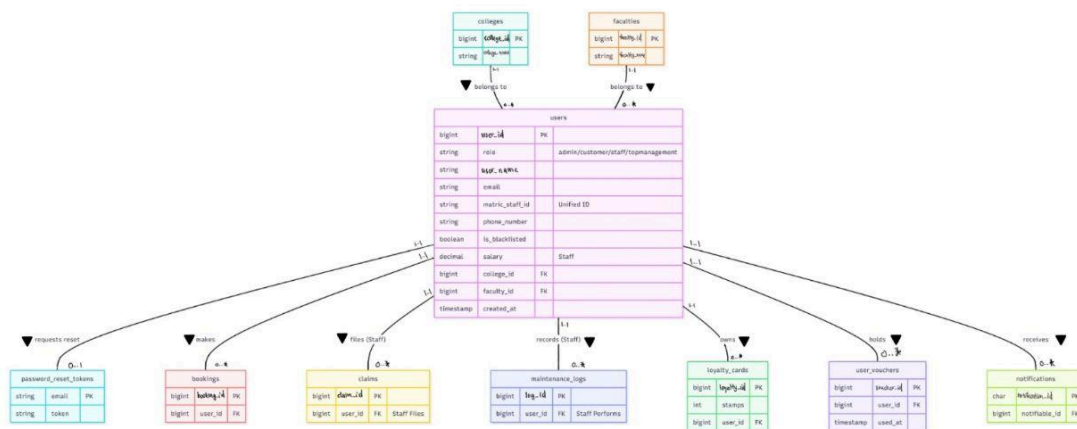


Figure 9.1.5.1 User Management Local ERD

9.2 Module 2: Vehicle Module

9.2.1 Module Description

The Vehicle Module is responsible for managing all information related to the cars available for rental. It allows staff to input and maintain details such as the vehicle's model, capacity, price per hour, and availability. Users can view a list of available cars based on their rental preferences (e.g., date, time).

When a booking is confirmed or rejected, the vehicle's availability status is automatically updated in the system. The module also allows staff to manage maintenance records, track road tax and insurance details for each vehicle. Additionally, an alert system is in place to notify staff when a vehicle's documents (e.g., insurance or road tax) are nearing expiration.

9.2.2 Transaction Requirement: Data Entry

1. **Add Vehicle:** Staff registers a new vehicle by entering details such as brand, model, year, capacity, fuel level, and base hourly rate. Crucially, they must also upload ownership validation documents including the Vehicle Grant (Geran), Insurance Policy, and Owner's IC for compliance.
2. **Update Vehicle Information:** Staff can update vehicle details, including availability status and maintenance records.
3. **Log Maintenance:** Staff can create maintenance records for a specific vehicle, inputting the service date, description of work, and associated costs. This action automatically updates the vehicle's status to "Maintenance".

9.2.3 Transaction Requirement: Data Update/Deletion

1. **Update Vehicle Information:** Staff can modify vehicle details including its pricing tier, images, and renew document expiry dates (Road Tax/Insurance). Updating valid expiry dates automatically restores a vehicle's status from "Unavailable" to "Available".
2. **Delete Vehicle:** Staff can permanently remove a vehicle profile from the system if it is sold or no longer part of the fleet, ensuring the booking system only displays currently operational vehicles.

9.2.4 Transaction Requirement: Data Queries

1. **List of Available Vehicles:** Search Available Vehicles: A query that filters vehicles based on a user's selected Pick-up and Return dates, excluding any vehicles that have conflicting bookings or are currently under maintenance/unavailable.

2. Vehicles with Expiring Documents: Query to display vehicles with expiring road tax or insurance within 30 days.
3. Booking Status: Query to display current booking statuses for all vehicles (e.g., confirmed, pending, rejected).
4. Vehicle Booking History: A query to retrieve the complete schedule of a specific vehicle, displaying past, active, and upcoming bookings to help staff visualize utilization.

9.2.5 Local Conceptual Data Model (ERD)

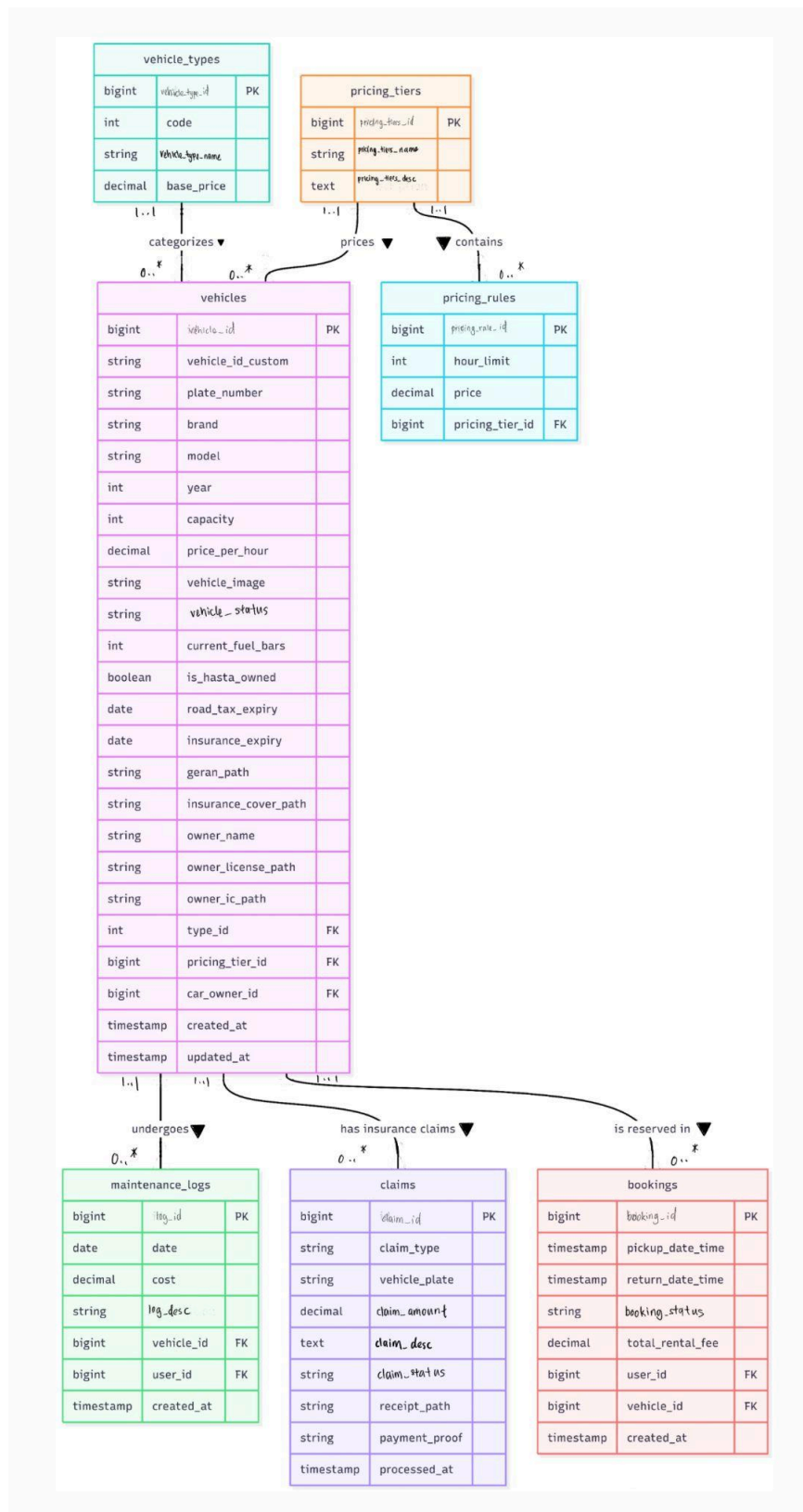


Figure 9.2.5.1 Vehicle Local ERD

9.3 Module 3: Booking Module

9.3.1 Module Description

The Booking Module is designed to handle the car rental process, allowing users to browse available vehicles based on pickup and drop-off times. Customers can select vehicles, specify the rental duration (enforced minimum of 1 hour), and book cars. The system applies additional charges for non-standard drop-off locations.

During the return process, users will be required to upload photos and provide the fuel level of the car. Automated email updates will be sent to customers regarding their booking status, ensuring transparency and real-time updates on confirmations, cancellations, and status changes.

9.3.2 Transaction Requirement: Data Entry

1. Booking Request: Customers input pickup date/time, return date/time, pickup location (e.g., UTM Office, Student Mall), and selecting a vehicle. They must also provide emergency contact details and refund bank information during this step.
2. Return Details: Users must upload photos of the vehicle and indicate the fuel level upon returning the vehicle.

9.3.3 Transaction Requirement: Data Update/Deletion

1. Update Booking: Customers can modify the time, location, or vehicle of a "Confirmed" booking, provided the new slot is available and the rental has not yet started.
2. Cancel Booking: Users can cancel their booking up to 24 hours before the scheduled pickup time. The system will then release the vehicle slot back to the "Available" pool and process any necessary voucher/stamp refund.

9.3.4 Transaction Requirement: Data Queries

1. Available Vehicles: Query to display available vehicles based on selected rental dates and times.
2. Booking History: Query to display previous bookings by the customer, including booking status and vehicle details.
3. Active Booking Status: A query to retrieve the current status of a specific booking (e.g., 'Waiting for Verification', 'Approved', 'Rejected') for the user dashboard.

9.3.5 Local Conceptual Data Model (ERD)

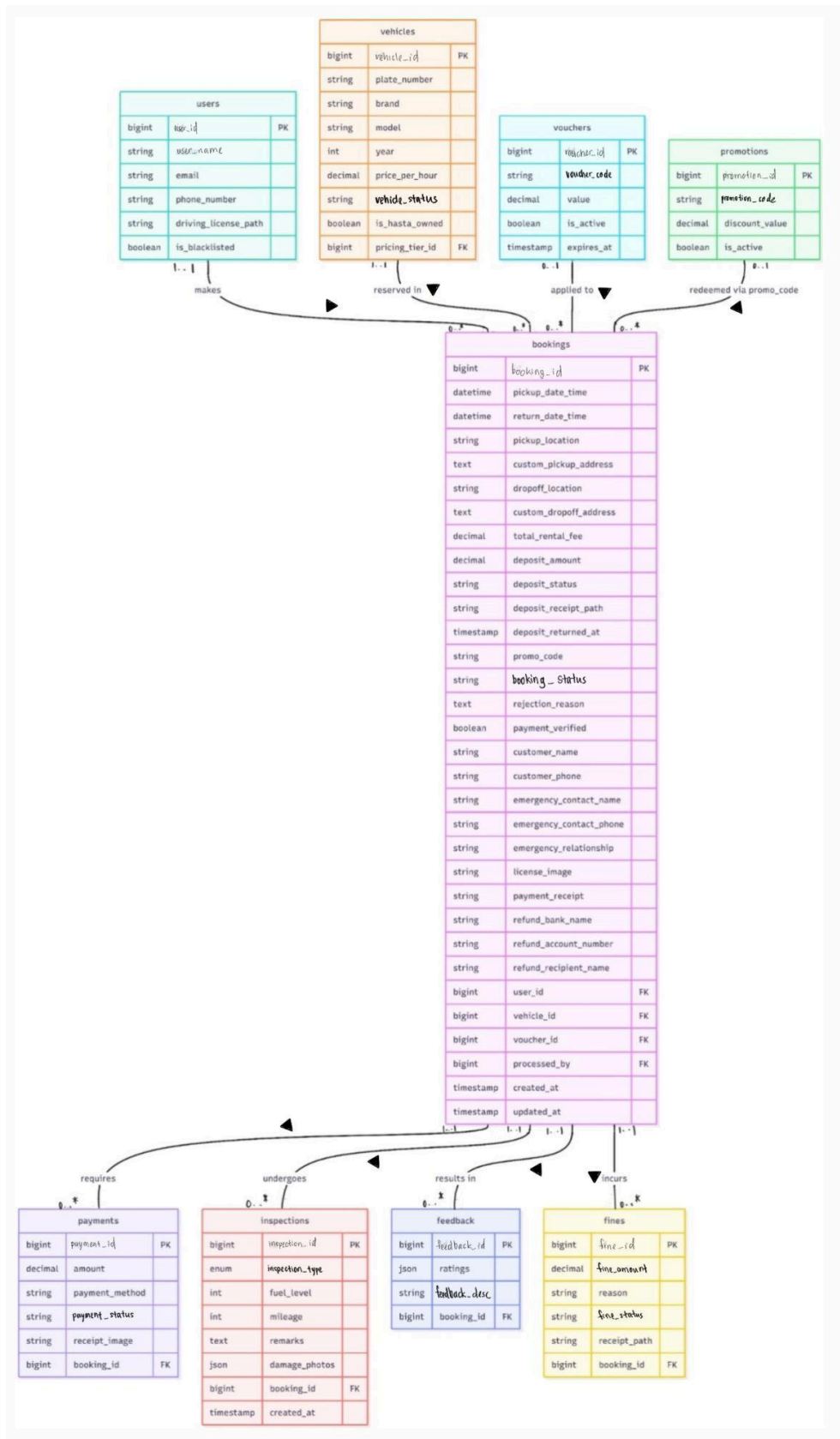


Figure 9.3.5.1 Booking Local ERD

9.4 Module 4: Payment Module

9.4.1 Module Description

The Payment Module facilitates the payment process for rentals through QR code-based transactions. Customers must upload their payment receipt, which is manually verified by staff. The system integrates a loyalty reward system where customers automatically earn a digital loyalty stamp for every completed rental exceeding 3 hours. These stamps accumulate to unlock tiered rewards, such as percentage discounts (e.g., 10% off at 3 stamps) or free rental hours.

9.4.2 Transaction Requirement: Data Entry

1. Payment Details: Customers input their payment details and upload the payment receipt for verification.
2. Process Refund: Admin staff input the "Refund Receipt" file and transaction details when returning the deposit to the customer.

9.4.3 Transaction Requirement: Data Update/Deletion

1. Update Payment Status: Once payment is verified, update the booking status to confirm.
2. Refund Process: When a booking is canceled, the refund status is updated accordingly.
3. Loyalty Adjustments: The system automatically increments a user's stamp count upon booking completion and revokes stamps if a booking is cancelled.

9.4.4 Transaction Requirement: Data Queries

1. Booking Payments: Query to retrieve payment information for all confirmed bookings.
2. Pending Verifications: A query to list all bookings with proper receipt uploads that are awaiting staff approval.
3. Loyalty Status: A query to retrieve a user's current stamp count and check their eligibility for specific reward tiers.
4. Financial Reports: Queries that aggregate total revenue, fines collected, and deposits returned for financial auditing.

9.4.5 Local Conceptual Data Model (ERD)

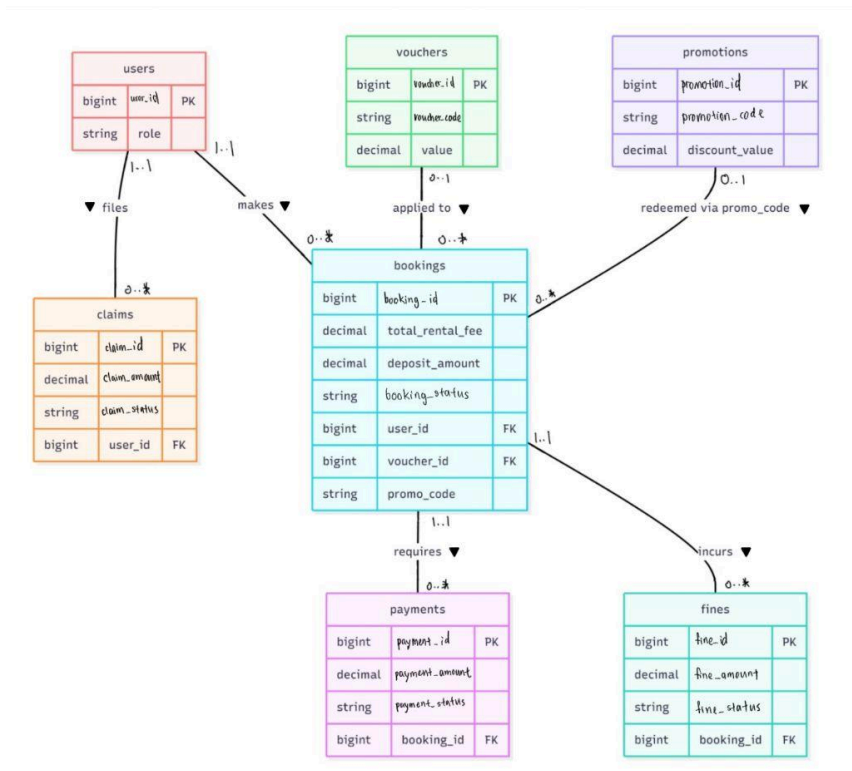


Figure 9.4.5.1 Payment Local ERD

9.5 Module 5: Reporting Module

9.5.1 Module Description

The Reporting Module generates both financial and booking reports. It helps management track daily, weekly, and monthly booking trends, revenue, and customer engagement. The system visualises booking data by different colleges and faculties to analyse student engagement and usage patterns, aiding in business strategy and operational improvements.

9.5.2 Transaction Requirement: Data Entry

1. Custom Report Filters: Users can input filters like date range, or report type to generate custom reports.

9.5.3 Transaction Requirement: Data Update/Deletion

1. Refresh Report Data: Since reports are generated dynamically, updating a source record (e.g., approving a pending claim or verifying a payment) automatically updates the relevant figures in all financial reports.

9.5.4 Transaction Requirement: Data Queries

1. Generate Financial Reports: Query to generate financial reports for specific time periods, tracking revenue, expenses, and profit.
2. Customer Engagement: A query that counts and groups active customers by their registered Faculty and College, helping the business understand its primary user base.
3. Net Profit Calculation: A complex query that aggregates the sum of all verified rental payments and paid fines, then subtracts total active staff salaries and approved claims to derive the current Net Profit.
4. Revenue Trend Analysis: A query that groups successful bookings by date (day/week/month) to visualize income trajectory and compare it against previous periods.

10.0 Benefit and Summary of Proposed System

The proposed "Hasta Car Rental" system transforms the traditional car rental process into a streamlined, web-based platform. By digitizing core operations from booking to vehicle return, the system enhances operational efficiency, data accuracy, and user satisfaction.

10.1 Benefits of the Proposed System

1. Enhanced Operational Efficiency

The system automates distinct processes such as price calculation, booking availability checks, and loyalty stamp assignment. This significantly reduces manual errors and administrative workload.

The system prevents double-booking by enforcing strict validation checks on vehicle availability during the reservation process.

The system also consolidates customer profiles, vehicle fleets and financial records into a single database, making information retrieval instant and reliable.

2. Customer Engagement

The system features a built-in reward system that tracks rental history and automatically awards "Stamps" which convert into redeemable vouchers, encouraging repeat bookings.

Customers have access to their booking history, current rental status and transparent pricing breakdowns, fostering customer trust.

3. Better Fleet Management & Accountability

Staff can record vehicle conditions (before and after rental) with photo evidence and damage logging directly in the system, resolving disputes effectively.

Staff can also log vehicle maintenance and inspection history to ensure fleet safety and reliability. Road tax and insurance management prevents staff from renting the vehicle with expired documents to customers.

4. Financial Integrity

The system allows users to track payments, deposits, fines, refunds and claims within the system, providing a clear financial audit trail.

10.2 Summary of System Modules

The system is composed of several key interconnected modules:

1. User Module: Identification & Verification

This module serves as the gateway for users, using matric/staff IDs for secure access. To ensure security and compliance, the module implements mandatory verification of NRIC/Passport, driving license and matric/staff ID. This module also supports profile updates and account deletions for privacy. Additionally, this module includes a Blacklist System and a Loyalty System to track rental history and stamps.

2. Vehicle Module: Fleet Management

Staff use this module to maintain the rental inventory and ensure vehicle safety.

This module supports management of vehicle specifications and availability. This module also allows users to track maintenance logs and provides 30-day alerts for expiring road tax and insurance. This module automatically updates vehicle availability based on booking outcomes.

3. Booking Module: Rental Lifecycle

This module handles the core transaction process between the customer and the fleet.

This module enables searching by date and time with a minimum 1-hour rental policy and surcharges for non-standard drop-off locations.

This module also requires users to upload vehicle photos and fuel levels upon pickup and return of vehicle to ensure accountability.

This module features automated email notifications to keep users updated on booking status.

4. Payment Module: Transactions and Rewards

This module focuses on secure financial processing and customer retention.

The module supports transactions via QR code, but requires staff to manually verify the uploaded receipt before confirming a booking. The module automatically awards loyalty stamps for any rental exceeding 3 hours, which can be redeemed for future rewards. This module also manages the refunds for completed or cancelled booking.

5. Reporting Module: Analytics & Strategy

This module is designed for management to monitor business health and user behaviour.

This module featured report generation on revenue, expenses, and profit over custom timeframes. Additionally, this module helps managers identify booking trends and

vehicle usage patterns. This module also visualizes engagement data based on UTM colleges and faculties to help tailor marketing and operational strategies.

11.0 Summary

To conclude, the proposed Car Rental System solves main limitations of the current manual processes used by Hasta Travel & Tours. By integrating booking, vehicle, customer, and transaction records into a single system with a centralized database, daily operations can be managed in a more organized and consistent manner. This reduces repetitive manual work and improves the accuracy of stored data.

The use of a centralised database ensures that booking records, vehicle information, financial transactions, and customer data are stored consistently and kept up to date. In addition, the reporting features allow top management to review company financial performance and customer usage patterns. This helps top management make better decision-making. Role-based access control also helps protect sensitive data by restricting access according to staff responsibilities.

Overall, the proposed system provides a more reliable and structured platform for managing car rental operations at UTM. Features such as real-time vehicle availability, online booking, digital records and loyalty tracking help improve both staff efficiency and customer experience. As a result, the system supports smoother operations and better data management for Hasta Travel & Tours.

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